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Questions

Q1.

An electrophile

- ☒ **A** accepts a pair of electrons
- ☐ **B** always has a negative charge
- ☐ **C** always has a positive charge
- ☐ **D** donates a pair of electrons

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(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What type of bond breaking occurs in this process?

$$H_2O \rightarrow H^+ + OH^-$$

- ☐ **A** electrophilic
- ☐ **B** heterolytic
- ☐ **C** homolytic
- ☐ **D** ionic

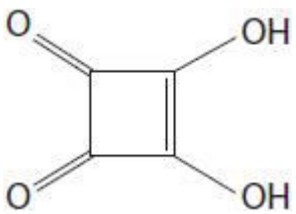
(Total for question = 1 mark)

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Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the molecular formula of the compound shown?



- ☐ **A** C_2HO_2

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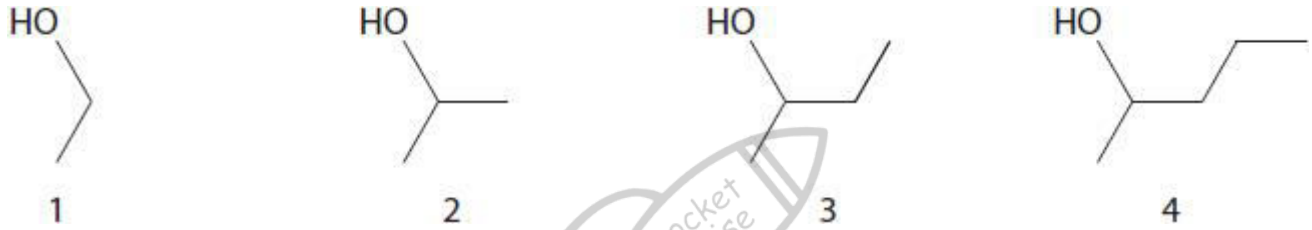
☒ **B** $C_4H_2O_4$
☒ **C** $C_4H_4O_4$
☒ **D** $C_4H_6O_4$

(Total for question = 1 mark)

Q4.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sequence of four molecules is shown.



(a) What is the molecular formula for the fifth molecule in this sequence?

(1)

☒ **A** CH_4O
☒ **B** $C_6H_{11}O$
☒ **C** $C_6H_{14}O$
☒ **D** $C_7H_{16}O$

(b) What is the name of this type of sequence?

(1)

☒ **A** functional group
☒ **B** homologous series
☒ **C** homolytic series
☒ **D** skeletal formulae

(Total for question = 2 marks)

Q5.

This question is about the compound ammonium dichromate(VI), $(NH_4)_2Cr_2O_7$.

(a) $(NH_4)_2Cr_2O_7$ breaks down at around 180 °C, producing an oxide of chromium, nitrogen and water as the only products.

(i) Write an equation for the reaction.
State symbols are not required.

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(2)

(ii) Give a name for this type of reaction.

(1)

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(b) When 0.00100 mol of ammonium dichromate(VI) was heated, 25.2 cm^3 of nitrogen gas, measured at laboratory temperature, was formed.

Calculate the temperature in the laboratory, in degrees Celsius, using the ideal gas equation. Use atmospheric pressure = 101 kPa.

$[pV = nRT \quad R = 8.31\ J\,mol^{-1}\,K^{-1}]$

(4)

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(c) (i) Draw a dot-and-cross diagram of the ammonium ion.

Use dots (●) for the nitrogen electrons and crosses (X) for the hydrogen electrons.

(2)

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(ii) Explain the shape of the ammonium ion using electron-pair repulsion theory.

(2)

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(d) Ammonium dichromate(VI) comes with several warnings on its packaging.

- This compound is toxic when inhaled and by passing through the skin.
- Handle with extreme caution.
- Contact can irritate and burn the skin and eyes, with possible eye damage.
- Inhaling can irritate the nose and throat.
- Ammonium dichromate(VI) is a strong oxidiser that enhances the combustion of other substances.

(i) Some of the symbols shown are used for ammonium dichromate(VI).
Identify the symbols for ammonium dichromate(VI) by placing a tick (✓) in the box under each relevant symbol.

(2)

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(ii) Suggest why ammonium dichromate(VI) is **not** stored in the same cupboard as alkanes.

(1)

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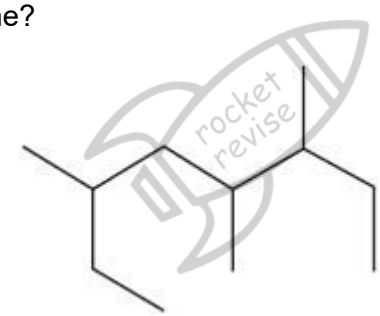
(Total for question = 14 marks)

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Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the IUPAC name of this alkane?



☒ **A** 2-ethyl-4,5-dimethylheptane

☒ **B** 6-ethyl-3,4-dimethylheptane

☒ **C** 3,4,6-trimethyloctane

☒ **D** 3,5,6-trimethyloctane

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(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A substance is labelled with the hazard symbol shown.

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What is the meaning of this symbol?

- ☒ **A** gloves must be worn
- ☒ **B** corrosive
- ☒ **C** do not store with flammable substances
- ☒ **D** oxidising

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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Heterolytic fission produces

- ☒ **A** free radicals only
- ☒ **B** ions only
- ☒ **C** free radicals and positive ions only
- ☒ **D** free radicals and negative ions only

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(Total for question = 1 mark)

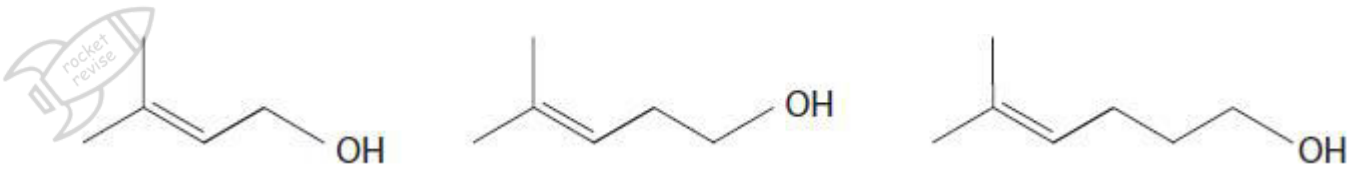
Q9.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the name of a series such as the one shown?



- ☒ **A** addition
- ☒ **B** heterolytic
- ☒ **C** homologous
- ☒ **D** homolytic

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

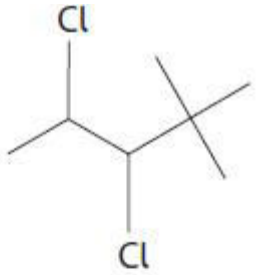
Which is the skeletal structure for 3,4-dichloro-2,2-dimethylpentane?

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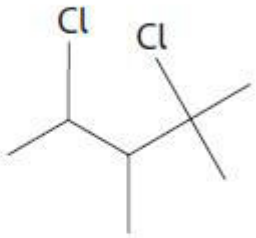
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☐ **A**

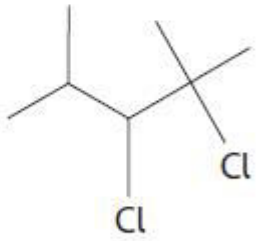


☐ **B**

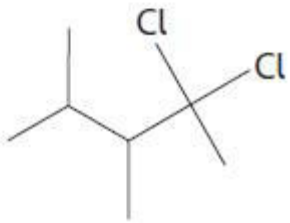


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☐ **C**



☐ **D**



(Total for question = 1 mark)

Q11.

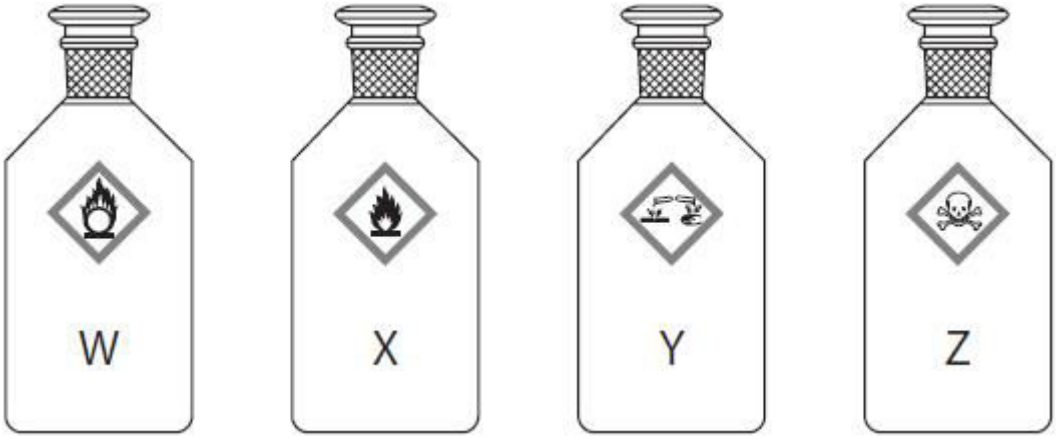
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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A technician is asked to store four reagent bottles safely. The only information about each substance is the hazard label on the bottle.

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Based on the hazard labels alone, which two bottles should **never** be stored together?

- ☒ **A** W and Z
- ☒ **B** Y and Z
- ☒ **C** W and X
- ☒ **D** Y and X

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(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Prop-2-en-1-ol is flammable, toxic and an environmental hazard.

Which hazard symbols should be shown on its container?

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☐ A 

☐ B 

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☐ C 

☐ D 

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(Total for question = 1 mark)

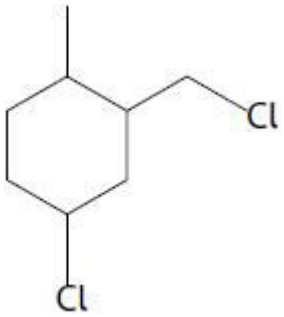
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Q13.

What is the empirical formula of the following molecule?



(1)

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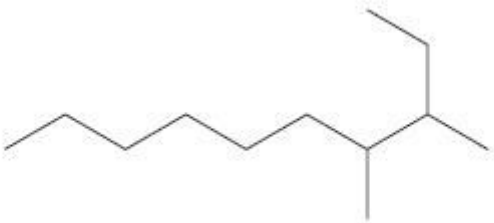
- ☒ **A** C_4H_4Cl
- ☒ **B** C_4H_7Cl
- ☒ **C** $C_8H_{11}Cl_2$
- ☒ **D** $C_8H_{14}Cl_2$

(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the IUPAC name for the hydrocarbon shown?



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- ☒ **A** 2-ethyl-3-methylnonane
- ☒ **B** 3,4-dimethyldecane
- ☒ **C** 8-ethyl-7-methylnonane
- ☒ **D** 7,8-dimethyldecane

(Total for question = 1 mark)

Q15.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which hazard warning symbol would be found on a bottle of hexane?

☐ A 

☐ B 

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☐ C 

☐ D 

(Total for question = 1 mark)

Q16.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) Hairdressers use solutions of hydrogen peroxide that have these hazards:

- can cause serious eye damage
- is harmful if swallowed
- is an oxidising agent

Which hazard symbols should be on containers of this solution?

(1)

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☒ **A**





☒ **B**





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☒ **C**





☒ **D**





(b) When hydrogen peroxide is used in an experiment to measure rate of decomposition, how could the risk be reduced?

(I) using an alternative method that involves less hazardous materials
(II) by working on a smaller scale
(III) taking precautions specific to the hazard

(1)

☒ **A** (I) only
☒ **B** (III) only
☒ **C** (II) and (III)
☒ **D** (I), (II) and (III)

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(Total for question = 2 marks)

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